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Temporal SCX-

——SpringRobbins-Monro

SCX

July 2, 2026

Abstract

SCXSpring SE-1——Robbins-Monro Spring SE-1[?] θ^***Temporal SCX**Spring SE-1 θ_t^* $\Omega(\sqrt{t})$—— [?]“” “” -$v_c = \Theta(\sigma/\sqrt{\eta})$—— $v < v_c\lambda = 1$ $v > v_c\lambda^* < 1$ $\lambda^* v$[?] t_{crit}—— $\gamma\tau$ $\gamma \cdot \tau \geq CC$ —— $\mathcal{F}_t\mathcal{M}$ $\ell_2\mathcal{F}_t$ “”

Robbins-Monro PL

Hilbert

Temporal SCXSpring SE-1- Robbins-Monro

1 SCX

1.1 SCX

SCXState CrystallizationSitus SpringYajieCercis Spring SE-1——$\mathcal{S}$Robbins-Monro

$$\theta_{t+1} = \theta_t - \alpha_t \cdot \nabla \mathcal{L}_{\mathrm{Spring}}(\theta_t; \mathcal{B}_t) + \xi_t, \tag{1}$$

α_tRobbins-Monro$\sum \alpha_t = \infty \sum \alpha_t^2 < \infty$ ξ_t

SCX——Theorem 1——

(1) “”$\Delta_s(t)$ ——

(2) Spring SE-1 θ^*——

(3) **Theorem 1**Chernoff-Hoeffding$F_1 \geq 1 - \frac{1}{\eta} \sum_s \rho_s \exp(-2M\Delta_s^2)$ ——distribution shift Δ_s

concept drift distribution shiftenvironmental non-stationarity

1.2

$tP_t(x_t, y_t)$ P_t——$\theta_t^* = \arg\min_\theta \mathbb{E}_{(x,y)\sim P_t}[\ell(x, y; \theta)]$

(1) $\theta_s^* s \ll t$ θ_t^*

(2)

“”____ - “”“”

—— Spring SE-1

1.3 SCX

SCX **??**

Temporal SCX		SCX	
	$\leftrightarrow$		“”$\rightarrow$“”
-	$\leftrightarrow$		v_c vs t_{crit}
	$\leftrightarrow$	Theorem 1	- vs -
$\mathcal{F}_t$	$\leftrightarrow$	Spring SE-1	$\rightarrow$

Table 1: Temporal SCXSCX

1.4

“—”SCX

2 Spring SE-1

2.1 Spring SE-1

Spring SE-1[?] $\mathcal{L}(\theta)L$-Polyak-ŁojasiewiczPL $\mu > 0\theta$

$$\frac{1}{2}\|\nabla\mathcal{L}(\theta)\|^2 \geq \mu(\mathcal{L}(\theta) - \mathcal{L}(\theta^*)), \tag{2}$$

θ^*

Lemma 2.1 (Spring SE-11.1). *Robbins-Monro*$\alpha_t = \frac{2}{\mu(t+t_0)}$*Spring SE-1*

$$\mathbb{E}[\mathcal{L}(\theta_t) - \mathcal{L}(\theta^*)] \leq \frac{2L\sigma^2}{\mu^2} \cdot \frac{1}{t} + O\left(\frac{1}{t^2}\right), \tag{3}$$

$\sigma^2 = \mathbb{E}[\|\xi_t\|^2]$

Spring SE-1$O(1/t)$ θ^*——α_t “”

2.2

Definition 2.2. $\{\theta_t^*\}_{t=0}^{\infty}\mathbb{R}^d$

$$v_t = \left\|\theta_{t+1}^* - \theta_t^*\right\|, \qquad v = \limsup_{T\to\infty} \frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}[v_t] \tag{4}$$

v-$t\mathbb{E}[v_t] \leq v$

1: $\theta_{t+1}^* = \theta_t^* + \varepsilon_t$ $\varepsilon_t \sim \mathcal{N}(0, \nu^2 I_d)v = \nu\sqrt{d} \cdot \sqrt{2/\pi}$

2: $\theta_t^* = \theta_0^* + t \cdot v \cdot \mathbf{u}$ $\mathbf{u}$

“” “”

Definition 2.3 (Spring SE-1). ***Spring SE-1****Spring SE-1*θ_t^* $tP_t(x_t, y_t)$ $g_t = \nabla_\theta \ell(x_t, y_t; \theta_t)$

$$\theta_{t+1} = \theta_t - \alpha_t \cdot g_t \tag{5}$$

$\mathbb{E}[g_t \mid \theta_t, \theta_t^*] = \nabla\mathcal{L}(\theta_t; \theta_t^*)$ $\mathrm{Cov}(g_t \mid \theta_t, \theta_t^*) = \Sigma_t \mathrm{Tr}(\Sigma_t) \leq \sigma^2$

Spring SE-1“θ^*”θ^* $\mathbb{E}[\|\theta_t - \theta_t^*\|^2]$ -

2.3

Spring SE-1“”θ_t——θ_t

Definition 2.4. $w : \mathbb{N} \times \mathbb{N} \to [0,1] w(t,s)$ $sts \leq t\theta_t$

$$\theta_t = \theta_0 - \sum_{s=0}^{t-1} \alpha_s \cdot w(t,s) \cdot g_s, \tag{6}$$

$\sum_{s=0}^{t-1} w(t,s) = 1$

(1) **Monotonic Memory, MM**$w(t,s) = 1/t$ Spring SE-1——

(2) **Exponential Forgetting, EF** $w(t,s) = \frac{(1-\gamma)^{t-s}}{\sum_{k=0}^{t-1}(1-\gamma)^k}$ $\gamma \in (0,1]$

(3) **Finite Window, FW** $w(t,s) = 1/W \cdot \mathbf{1}[t-s \leq W]$ $W \in \mathbb{N}W$

Remark 2.5. $\gamma = 1/W$ $1/\gamma W$——

3

3.1

Theorem 3.1 (——)**.** $\theta_t^* \liminf_{t\to\infty} \mathbb{E}[\left\|\theta_{t+1}^* - \theta_t^*\right\|] \geq v_{\min} > 0$ $w(t,s) = 1/t$ $\alpha_t \sum \alpha_t = \infty \sum \alpha_t^2 < \infty$

$$\boxed{\lim_{t\to\infty} \mathbb{E}[\|\theta_t - \theta_t^*\|^2] = \infty} \tag{7}$$

Spring SE-1 ——

Corollary 3.2. **??**$v_{\min} > 0$

$$\mathbb{E}[\|\theta_t - \theta_t^*\|^2] = \Omega(v_{\min}^2 \cdot t) \tag{8}$$

$\theta_t^* = \theta_0^* + vt\mathbf{u}$

$$\mathbb{E}[\|\theta_t - \theta_t^*\|^2] \geq \frac{v^2 t^2}{4} + o(t^2) \tag{9}$$

3.2

Proof. **1**

$$\mathbb{E}[\|\theta_t - \theta_t^*\|^2] = \underbrace{\|\mathbb{E}[\theta_t] - \theta_t^*\|^2}_{2} + \underbrace{\mathbb{E}[\|\theta_t - \mathbb{E}[\theta_t]\|^2]} \tag{10}$$

2$w(t,s) = 1/t\theta_t$

$$\begin{aligned}\mathbb{E}[\theta_t] &= \theta_0 - \sum_{s=0}^{t-1} \frac{\alpha_s}{t} \cdot \mathbb{E}[g_s] \\ &= \theta_0 - \frac{1}{t}\sum_{s=0}^{t-1} \alpha_s \cdot \nabla\mathcal{L}(\mathbb{E}[\theta_s]; \theta_s^*)\end{aligned} \tag{11}$$

$\mathbb{E}[\theta_t]\theta_0^*, \theta_1^*, \ldots, \theta_{t-1}^*$ $\theta_t^* \theta_s^*$“” $\mathbb{E}[\theta_t]\theta_t^*$
3$\bar{\theta}_t^* = \frac{1}{t}\sum_{s=0}^{t-1} \theta_s^*$ $\nabla\mathcal{L}(\mathbb{E}[\theta_s]; \theta_s^*) \propto \mathbb{E}[\theta_s] - \theta_s^*$ $\mathbb{E}[\theta_t] \approx \bar{\theta}_t^*$——$\theta_t$

$$\begin{aligned}\|\mathbb{E}[\theta_t]-\theta_t^*\|^2 &\approx \left\|\bar{\theta}_t^*-\theta_t^*\right\|^2\\ &= \left\|\frac{1}{t}\sum_{s=0}^{t-1}(\theta_s^*-\theta_t^*)\right\|^2 \end{aligned} \tag{12}$$

4$v_{\min} > 0$ $\|\theta_s^* - \theta_t^*\| \geq v_{\min} \cdot (t-s) - o(t-s)$

$$\begin{aligned}\left\|\bar{\theta}_t^*-\theta_t^*\right\|^2 &\geq \left(\frac{1}{t}\sum_{s=0}^{t-1} v_{\min}(t-s)\right)^2 - o(t^2)\\ &= v_{\min}^2 \cdot \left(\frac{t(t+1)}{2t}\right)^2 - o(t^2)\\ &= \frac{v_{\min}^2 t^2}{4} + o(t^2)\end{aligned} \tag{13}$$

5Robbins-Monro $\mathbb{E}[\|\theta_t - \mathbb{E}[\theta_t]\|^2] = O(\sum_{s=0}^{t-1}\alpha_s^2 \cdot w(t,s)^2) = O(1/t)$ $\sum \alpha_s^2 < \infty w(t,s) = 1/t$
6$^2 = \Omega(t^2)$= $O(1/t)$ $\mathbb{E}[\|\theta_t - \theta_t^*\|^2] = \Omega(t^2) \to \infty$ □

[] **1**–**5**3≈ $\ell(x,y;\theta) = \frac{1}{2}\|y - f(x;\theta)\|^2 f$ $\mathbb{E}[\theta_t]\bar{\theta}_t^*$ ——Robbins-Monro ——$\lim_{t\to\infty} \mathbb{E}[\|\theta_t - \theta_t^*\|^2] = \infty$ $v_{\min}t$ Ornstein-Uhlenbeck —— §7

3.3

??

Table 2:

	$v_{\min} > 0$
$\Delta_s(t)$	$\mathbb{E}[\|\theta_t - \theta_t^*\|^2]$
$\Delta_s \to 0$	$\|\theta_t - \theta_t^*\|^2 \to \infty$
$O((1-\gamma)^\tau)$	$\Omega(v_{\min}^2 t^2)$
$\to \Delta_s > 0$	$\to$

“” “”____ “”“”

3.4

Corollary 3.3. *SCX*

(1)

(2) data staleness

(3)

——

[] “” “” “”____

4 -

4.1

——

Theorem 4.1 (-)**.** $w(t,s;\lambda) \propto \lambda^{t-s}\lambda \in (0,1]$ $\lambda = 1\lambda \to 0$

$$\mathrm{MSE}(\lambda) = \limsup_{t\to\infty} \mathbb{E}[\|\theta_t - \theta_t^*\|^2] \tag{14}$$

PL(**??**)$\alpha_t = \alpha$ $\sigma^2 v$

$$\boxed{v_c = \frac{\sigma}{2\sqrt{\alpha}} \cdot \sqrt{\frac{\mu}{L}}} \tag{15}$$

(1) $v < v_c \mathrm{MSE}(\lambda)\lambda = 1$——

(2) $v > v_c \mathrm{MSE}(\lambda)\lambda^* < 1$

$$\lambda^* = 1 - \frac{\alpha\mu}{2L} \cdot \left(\sqrt{1 + \frac{8Lv^2}{\alpha\mu\sigma^2}} - 1\right) \tag{16}$$

$\lambda^* v \lim_{v\to\infty} \lambda^* = 0$

(3) $v = v_c \mathrm{MSE}(\lambda)\lambda = 1$——

Figure 1: -$\lambda\lambda = 1$ $\mathrm{MSE}(\lambda) v < v_c \mathrm{MSE}\lambda = 1$ $v > v_c \mathrm{MSE}\lambda^* < 1\star$ $v = v_c \mathrm{MSE}\lambda = 1$

4.2

Proof. **1**αSpring SE-1

$$\theta_{t+1} = \theta_t - \alpha \cdot g_t \tag{17}$$

$w(t,s;\lambda) \propto \lambda^{t-s}$ $O(\lambda^t)$

$$\theta_t \approx -\alpha \sum_{s=-\infty}^{t-1} \lambda^{t-s} \cdot g_s \cdot (1-\lambda) \tag{18}$$

$(1-\lambda)1$

2-PL $g_t g_t \approx H(\theta_t - \theta_t^*) + \xi_t H = \nabla^2 \mathcal{L}(\theta^*)$ $\mu I \preceq H \preceq LI$
$t \to \infty \theta_t$

$$b(\lambda) = \|\mathbb{E}_\infty[\theta_t] - \theta_t^*\| \quad , \tag{19}$$

$$\sigma^2(\lambda) = \mathrm{Tr}(\mathrm{Cov}_\infty(\theta_t)) \tag{20}$$

3$\theta_t^* = \theta_0^* + vt\mathbf{u}$ $\mathbb{E}_\infty[\theta_t]\theta_t^*$

$$\begin{aligned}
\mathbb{E}_\infty[\theta_t] &= (1-\lambda) \sum_{k=0}^{\infty} \lambda^k \cdot \mathbb{E}_\infty[\theta_{t-k} - \alpha g_{t-k-1}] \\
&\approx (1-\lambda) \sum_{k=0}^{\infty} \lambda^k \cdot (\theta_{t-k}^* - \alpha H(\mathbb{E}_\infty[\theta_{t-k}] - \theta_{t-k}^*))
\end{aligned} \tag{21}$$

vPL$\alpha\mu \ll 1$

$$b(\lambda)^2 = \frac{v^2}{\alpha^2\mu^2} \cdot \left(\frac{\lambda}{1-\lambda}\right)^2 + o\left(\frac{1}{(1-\lambda)^2}\right) \tag{22}$$

$\lambda \to 1$$1/(1-\lambda)^2$—— $1/(1-\lambda)$$\lambda \to 0$ $v^2/(\alpha^2\mu^2)$——

4

$$\begin{aligned}\sigma^2(\lambda) &= \alpha^2 \cdot (1-\lambda)^2 \sum_{k=0}^{\infty} \lambda^{2k} \cdot \mathrm{Tr}(\Sigma_\infty) \\ &= \alpha^2\sigma^2 \cdot \frac{(1-\lambda)^2}{1-\lambda^2} \\ &= \alpha^2\sigma^2 \cdot \frac{1-\lambda}{1+\lambda}\end{aligned} \tag{23}$$

$\lambda \to 1$$\alpha^2\sigma^2 \cdot 0 = 0$ ——$\lambda \to 1$$(1-\lambda)/(1+\lambda) \to 0$ $t$$O(1/t)$ $\lambda \to 0$$\alpha^2\sigma^2$——

5MSE

$$\mathrm{MSE}(\lambda) = \frac{v^2}{\alpha^2\mu^2} \cdot \frac{\lambda^2}{(1-\lambda)^2} + \alpha^2\sigma^2 \cdot \frac{1-\lambda}{1+\lambda} \tag{24}$$

λ

$$\frac{\mathrm{d}\,\mathrm{MSE}(\lambda)}{\mathrm{d}\lambda} = 0 \Longrightarrow \frac{2v^2}{\alpha^2\mu^2} \cdot \frac{\lambda}{(1-\lambda)^3} - \alpha^2\sigma^2 \cdot \frac{2}{(1+\lambda)^2} = 0 \tag{25}$$

$$\frac{v^2}{\mu^2} \cdot \frac{\lambda(1+\lambda)^2}{(1-\lambda)^3} = \alpha^4\sigma^2 \tag{26}$$

6$\lambda \to 1$ $\lambda = 1-\varepsilon$$\varepsilon \to 0^+$$\lambda(1+\lambda)^2/(1-\lambda)^3 \approx 4/\varepsilon^3$ (**??**)

$$\frac{4v^2}{\mu^2\varepsilon^3} \approx \alpha^4\sigma^2 \Longrightarrow \varepsilon \approx \left(\frac{4v^2}{\alpha^4\mu^2\sigma^2}\right)^{1/3} \tag{27}$$

$v$$\varepsilon \ll 1$$\lambda^* \approx 1-\varepsilon$1 $v$$\varepsilon$$\lambda^*$$\lambda^* = 1$$\varepsilon = 0$ $v = 0$

v_cMSE(1)$v$$\frac{\mathrm{d}\,\mathrm{MSE}}{\mathrm{d}\lambda}\big|_{\lambda=1}$

$$\frac{\mathrm{d}\,\mathrm{MSE}}{\mathrm{d}\lambda}\Big|_{\lambda=1} = \lim_{\lambda \to 1^-}\left[\frac{2v^2}{\alpha^2\mu^2} \cdot \frac{\lambda}{(1-\lambda)^3} - \alpha^2\sigma^2 \cdot \frac{2}{(1+\lambda)^2}\right] \tag{28}$$

$v > 0$$\to +\infty$——$\lambda$ MSE$v > 0$$\lambda^* < 1$ "λ^*10"

MSE$\lambda = 1$$\lambda = \lambda^*$

$$\Delta\,\mathrm{MSE} = \frac{\mathrm{MSE}(1) - \mathrm{MSE}(\lambda^*)}{\mathrm{MSE}(1)} \tag{29}$$

$v_c$$\Delta\,\mathrm{MSE}$10%$v$

$$v_c = \Theta\left(\frac{\sigma}{\sqrt{\alpha}} \cdot \sqrt{\frac{\mu}{L}}\right) \tag{30}$$

7L-L (**??**)

$$\mathrm{MSE}(\lambda) = \frac{L^2v^2}{\mu^4} \cdot \frac{\lambda^2}{(1-\lambda)^2} + \frac{\alpha\sigma^2}{\mu} \cdot \frac{1-\lambda}{1+\lambda} \tag{31}$$

$\frac{\mathrm{d}\,\mathrm{MSE}}{\mathrm{d}\lambda}\big|_{\lambda=1} = 0$

$$v_c^2 = \frac{\alpha\sigma^2\mu^3}{4L^2} \tag{32}$$

$v_c = \frac{\sigma}{2\sqrt{\alpha}} \cdot \frac{\mu^{3/2}}{L}$ $\mu \approx L$$v_c \approx \frac{\sigma}{2\sqrt{\alpha}} \cdot \sqrt{\frac{\mu}{L}}$

□

[] **1–4-** 5–7$\lambda \to 1$——$1-\lambda \ll 1$ λ^*1

(1) PL—— PL

(2) $g_t \approx H(\theta_t - \theta_t^*) + \xi_t$ $\|\theta_t - \theta_t^*\|$—— Lyapunov §7

(3) α—— “”

4.3

-——

Table 3: -

	-
t	$v\lambda$
t_{crit}	$v_c/\ \lambda^*$
$\|\nabla\mathcal{L}\|$	MSEλ
PL$O(1/t)$	MSE$\lambda^* < 1$
$\|\nabla\mathcal{L}\| \leq \mu \cdot \mathrm{diam}$	$\frac{\mathrm{d\,MSE}}{\mathrm{d}\lambda}\big\|_{\lambda=1}$
“”“”	“”“”

—— - $\kappa = L/\mu\sigma^2$

Corollary 4.2. -“” $\gamma \in [0,1]\gamma = 0\gamma = 1$ v_c

[] MSE$(\gamma)\gamma = 0vv_c$ generic property

4.4

MSE$(\lambda)\lambda = 1$

(1) $v < v_c$——$\lambda < 1$1 MSE“”

(2) $v > v_c$——λ^*1 v

(3) **critical slowing down**$v \approx v_c$ MSE$(\lambda)\lambda = 1$—— v

[] $t \to \infty$ t“”crossover—— $t\bar{v}_t = \frac{1}{t}\sum_{s=0}^{t-1} v_s$ $v_c(t) = v_c \cdot (1 + O(1/\sqrt{t}))$

5 -

5.1

12 ——

(1) **Forgetting Speed**

(2) **Detection Delay**“”

——

5.2

Definition 5.1. $tW\{x_{t-W+1}, \ldots, x_t\}$

$$H_0 : P_t = P_{t-1} = \cdots = P_{t-W+1} \quad , \tag{33}$$
$$H_1 : P_t \neq P_{t-1} \quad t \tag{34}$$

$\tau 1 - \alpha$

Theorem 5.2 (-)**.** $\gamma = 1 - \lambda \in (0, 1]\delta$ *Wasserstein-2*$\mathcal{W}_2(P_t, P_{t-1}) \geq \delta$ τ

$$\tau \geq \frac{\sigma^2}{2\delta^2} \cdot \log \frac{1}{\alpha} \cdot \frac{1}{\gamma} - O(1) \tag{35}$$

$\gamma\tau$

$$\boxed{\gamma \cdot \tau \geq C(\delta, \sigma, \alpha)} \tag{36}$$

$C(\delta, \sigma, \alpha) = \frac{\sigma^2}{2\delta^2} \cdot \log \frac{1}{\alpha}$

——

5.3

Proof. **1**st $(1-\gamma)\gamma^{t-s}$“”ESS

$$N_{\text{eff}}(\gamma) = \frac{(\sum_{k=0}^{\infty}(1-\gamma)\gamma^k)^2}{\sum_{k=0}^{\infty}(1-\gamma)^2\gamma^{2k}} = \frac{1}{1-\gamma^2} \cdot (1-\gamma)^2 \cdot \frac{1}{(1-\gamma)^2/(1-\gamma^2)} = \frac{1+\gamma}{1-\gamma} \tag{37}$$

$N_{\text{eff}}(\gamma) \approx 2/\gamma\gamma \ll 1$——
2 dESS $= N_{\text{eff}}$ δLe CamChernoff-Stein

$$N_{\text{eff}} \geq \frac{\sigma^2}{2\delta^2} \cdot \log \frac{1}{\alpha} \cdot (1 + o(1)) \tag{38}$$

——LRT$N_{\text{eff}} = \frac{\sigma^2}{2\delta^2} \log \frac{1}{\alpha}$
3τN_{eff} $tN_{\text{eff}}(t) \approx \frac{1+\gamma}{1-\gamma} \cdot (1-\gamma^t)$ $t \gg 1/\gamma N_{\text{eff}}(t) \to (1+\gamma)/(1-\gamma) \approx 2/\gamma$
$N_{\text{eff}} \geq \frac{\sigma^2}{2\delta^2} \log \frac{1}{\alpha}$ τ

$$\frac{1+\gamma}{1-\gamma} \cdot (1-\gamma^\tau) \geq \frac{\sigma^2}{2\delta^2} \log \frac{1}{\alpha} \tag{39}$$

$\tau\gamma \ll 1$

$$\tau \geq \frac{\sigma^2}{2\delta^2} \cdot \log \frac{1}{\alpha} \cdot \frac{1}{\gamma} - O(1) \tag{40}$$

4 ——Chernoff-Stein $\gamma\tau < C/\gamma$ □

[] 1–4 Le CamChernoff-Stein

(1) $C(\alpha)/\delta^2$ $C(\alpha)$——$\gamma \cdot \tau = \Omega(1/\delta^2)$ Le Cam

(2) “”N_{eff} $N_{\text{eff}} = W$ $\tau/W \geq C/\delta^2$——

5.4

Definition 5.3. $\pi\pi'$

- $\tau(\pi') \leq \tau(\pi)$
- $\gamma(\pi') \leq \gamma(\pi)$
-

Pareto frontier

Corollary 5.4. $\{\pi_\gamma : \gamma \in (0,1]\}$-

$$\mathcal{P} = \left\{(\gamma, \tau) : \tau = \frac{C}{\gamma},\ \gamma \in (0,1]\right\} \tag{41}$$

$\{\pi_W : W \in \mathbb{N}\}$

$$\mathcal{P}_W = \left\{\left(\frac{1}{W}, \tau\right) : \tau = C \cdot W,\ W \in \mathbb{N}\right\} \tag{42}$$

“” ——(δ, σ, α)

[] $\gamma\lambda$ $\gamma\lambda$ ——

5.5 Theorem 1

SCX Theorem 1 $F_1 \geq 1 - \frac{1}{\eta}\sum_s \rho_s \exp(-2M\Delta_s^2)$—— MF_1

$\gamma\tau$

Theorem 1“” “” SCX—— §7

6 $\mathcal{F}_t$

6.1

Spring——

“”

6.2

Definition 6.1. $\mathcal{M}$*Hilbert*$m \in \mathcal{M}$ $\mathcal{M}\langle\cdot,\cdot\rangle_{\mathcal{M}}\|\cdot\|_{\mathcal{M}}$
$m_t t$ $\mathcal{M}d$*Hilbert*

Definition 6.2. $t\mathcal{I}_t : \mathcal{X} \times \mathcal{Y} \to \mathcal{M}$ (x_t, y_t)

$$\delta m_t = \mathcal{I}_t(x_t, y_t) \tag{43}$$

Spring SE-1$\delta m_t = -\alpha_t \cdot g_t$

Definition 6.3. $\mathcal{F}_t : \mathcal{M} \times \mathcal{M} \to \mathcal{M}$

$$m_{t+1} = \mathcal{F}_t(m_t, \delta m_t) \tag{44}$$

$m_t \delta m_t$

6.3

Axiom 6.4. *(A1)* ***Contractivity***$\rho_t \in [0,1]$ $m, m' \in \mathcal{M}\delta m$

$$\left\|\mathcal{F}_t(m, \delta m) - \mathcal{F}_t(m', \delta m)\right\|_{\mathcal{M}} \leq \rho_t \cdot \left\|m - m'\right\|_{\mathcal{M}} \tag{45}$$

ρ_t—— $\rho_t = 1\rho_t < 1$

(A2) ***Innovation Linearity*** $\mathcal{F}_t(m, \cdot)$

$$\mathcal{F}_t(m, a \cdot \delta m + b \cdot \delta m') = a \cdot \mathcal{F}_t(m, \delta m) + b \cdot \mathcal{F}_t(m, \delta m') - (a + b - 1) \cdot \mathcal{F}_t(m, 0) \tag{46}$$

$\mathcal{F}_t(m, 0) = m$ $\mathcal{F}_t(m, \delta m) = \rho_t m + (1 - \rho_t) \cdot \mathcal{T}_t(\delta m)$ $\mathcal{T}_t$“”

(A3) ***Causality***$\mathcal{F}_t\{m_s, \delta m_s\}_{s \leq t}$——

A1——“”“” A2 A3“”

6.4

Proposition 6.5. *A1–A3*

$$\boxed{\mathcal{F}_t(m, \delta m) = \rho_t \cdot m + (1 - \rho_t) \cdot \mathcal{T}_t(\delta m)} \tag{47}$$

$\rho_t \in [0,1]\mathcal{T}_t : \mathcal{M} \to \mathcal{M}$

Proof. A2$\mathcal{F}_t(m, 0) = \rho_t m$ δm

$$\begin{aligned} \mathcal{F}_t(m, \delta m) &= \mathcal{F}_t(m, 1 \cdot \delta m + 0 \cdot 0) \\ &= 1 \cdot \mathcal{F}_t(m, \delta m) + 0 \cdot \mathcal{F}_t(m, 0) - 0 \cdot \mathcal{F}_t(m, 0) \\ &= \rho_t \cdot m + (1 - \rho_t) \cdot \mathcal{T}_t(\delta m) \end{aligned} \tag{48}$$

$\mathcal{T}_t\mathcal{T}_t(\delta m) = \frac{\mathcal{F}_t(m,\delta m) - \rho_t m}{1 - \rho_t}$ $\rho_t < 1\rho_t = 1$ A1$\|\mathcal{T}_t(\delta m)\|_{\mathcal{M}} \leq \|\delta m\|_{\mathcal{M}}$ $\mathcal{T}_t$ □

——ρ_t $(1 - \rho_t)\gamma_t = 1 - \rho_t$

6.5

(**??**)

(1) $\rho_t = 1t$—— **??**

(2) **EF**$\rho_t = \lambda\mathcal{T}_t(\delta m) = \delta m$

$$m_{t+1} = \lambda m_t + (1 - \lambda)\delta m_t \tag{49}$$

(3) $\rho_t = \rho(\|\delta m_t\|, \|m_t\|, t)$——

ρ_t

Remark 6.6. FWA2 ——$\rho_t = 1 - \frac{1}{W} \cdot \sigma(t - t_0)$ σsigmoid$W \to \infty$

6.6

Theorem 6.7. $\{\mathcal{F}_t\}_{t=0}^{\infty}$ $\mathcal{F}_t(m, \delta m) = \rho_t m + (1-\rho_t)\mathcal{T}_t(\delta m)$ $\rho_t \in [0,1]\mathcal{T}_t$ $\{\delta m_t\}$ $\bar{\delta m} = \lim_{T\to\infty} \frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}[\delta m_t]$

(1) $\limsup_{t\to\infty} \rho_t \le \bar{\rho} < 1$ $\{m_t\}$

$$\left\|\mathbb{E}[m_\infty] - \bar{\delta m}\right\|_{\mathcal{M}} \le \frac{\bar{\rho}}{1-\bar{\rho}} \cdot \sup_t \left\|\mathbb{E}[\delta m_t] - \bar{\delta m}\right\|_{\mathcal{M}} \tag{50}$$

(2) $m_0 \mathbb{E}[m_t] O(\bar{\rho}^t)$

(3) $\{\delta m_t\} \mathrm{Cov}(\delta m_t, \delta m_s) = 0 t \ne s$

$$\mathbb{V}[m_\infty] = (1-\bar{\rho})^2 \sum_{k=0}^{\infty} \bar{\rho}^{2k} \cdot \mathbb{V}[\mathcal{T}_{t-k}(\delta m_{t-k})] = \frac{1-\bar{\rho}}{1+\bar{\rho}} \cdot \bar{\sigma}^2 \tag{51}$$

$\bar{\sigma}^2 = \limsup_t \mathbb{V}[\mathcal{T}_t(\delta m_t)]$

(4) **-**$\bar{\rho} \to 1$ $\bar{\rho} \to 1$ $\bar{\rho}^*$—— **??**

Proof. **1**$m_0 t$

$$\begin{aligned} m_t &= \rho_{t-1} m_{t-1} + (1-\rho_{t-1})\mathcal{T}_{t-1}(\delta m_{t-1}) \\ &= \left(\prod_{k=0}^{t-1} \rho_k\right) m_0 + \sum_{s=0}^{t-1} (1-\rho_s) \left(\prod_{k=s+1}^{t-1} \rho_k\right) \mathcal{T}_s(\delta m_s) \end{aligned} \tag{52}$$

2$\mathbb{E}[\delta m_t] \to \bar{\delta m}$

$$\begin{aligned} \mathbb{E}[m_t] &= \left(\prod_{k=0}^{t-1} \rho_k\right) m_0 + \sum_{s=0}^{t-1} (1-\rho_s) \left(\prod_{k=s+1}^{t-1} \rho_k\right) \mathbb{E}[\mathcal{T}_s(\delta m_s)] \\ &\to \frac{1-\bar{\rho}}{1-\bar{\rho}} \cdot \bar{\delta m} = \bar{\delta m} \quad \rho_t \equiv \bar{\rho} \end{aligned} \tag{53}$$

3

$$\begin{aligned} \|\mathbb{E}[m_\infty] - \mathbb{E}[\delta m_t]\|_{\mathcal{M}} &\le \sum_{k=0}^{\infty} (1-\bar{\rho})\bar{\rho}^k \cdot \|\mathbb{E}[\delta m_{t-k}] - \mathbb{E}[\delta m_t]\|_{\mathcal{M}} \\ &\le \frac{\bar{\rho}}{1-\bar{\rho}} \cdot \sup_s \left\|\mathbb{E}[\delta m_s] - \bar{\delta m}\right\|_{\mathcal{M}} \end{aligned} \tag{54}$$

4

$$\begin{aligned} \mathbb{V}[m_\infty] &= \mathbb{V}\left[(1-\bar{\rho}) \sum_{k=0}^{\infty} \bar{\rho}^k \mathcal{T}_{t-k}(\delta m_{t-k})\right] \\ &= (1-\bar{\rho})^2 \sum_{k=0}^{\infty} \bar{\rho}^{2k} \cdot \mathbb{V}[\mathcal{T}_{t-k}(\delta m_{t-k})] \\ &= (1-\bar{\rho})^2 \cdot \frac{\bar{\sigma}^2}{1-\bar{\rho}^2} = \frac{1-\bar{\rho}}{1+\bar{\rho}} \cdot \bar{\sigma}^2 \end{aligned} \tag{55}$$

□

[] $\rho_t \equiv \bar{\rho}$ 1–4

(1) $\bar{\rho}$—— Lyapunov§7 $\mathbb{E}[\log \rho_t] < 0$

(2) ——$\{\delta m_t\}$ AR(1)$\mathrm{Cov}(\delta m_t, \delta m_{t-k}) = \phi^k \bar{\sigma}^2$ $\mathbb{V}[m_\infty] = \frac{1-\bar{\rho}}{1+\bar{\rho}} \cdot \frac{1+\bar{\rho}\phi}{1-\bar{\rho}\phi} \cdot \bar{\sigma}^2$

6.7

$\mathcal{F}_t = \mathcal{F}t$

Proposition 6.8. $\{\mathcal{F}_\lambda : \lambda \in [0,1]\}$

$$\mathcal{F}_{\lambda_1} \circ \mathcal{F}_{\lambda_2} = \mathcal{F}_{\lambda_1\lambda_2} \tag{56}$$

$\mathcal{F}_1\mathcal{F}_0$

$$\mathcal{L}[m] = \lim_{\gamma\to 0^+} \frac{\mathcal{F}_{1-\gamma}[m] - m}{\gamma} = \mathcal{T}(\delta m) - m \tag{57}$$

$\mathcal{F}_\lambda = \exp((1-\lambda)\mathcal{L})$

Proof.

$$\begin{aligned}\mathcal{F}_{\lambda_1}(\mathcal{F}_{\lambda_2}(m, \delta m_1), \delta m_2) &= \lambda_1(\lambda_2 m + (1-\lambda_2)\delta m_1) + (1-\lambda_1)\delta m_2 \\ &= \lambda_1\lambda_2 m + \lambda_1(1-\lambda_2)\delta m_1 + (1-\lambda_1)\delta m_2\end{aligned} \tag{58}$$

$\delta m_1 = \delta m_2 = \delta m$ $\mathcal{F}_{\lambda_1} \circ \mathcal{F}_{\lambda_2}(m, \delta m) = \lambda_1\lambda_2 m + (1-\lambda_1\lambda_2)\delta m = \mathcal{F}_{\lambda_1\lambda_2}(m, \delta m)$

$\square$

$n\lambda\lambda^n$——

6.8

- ??

(1) $\mathcal{F}_\lambda\mu_\lambda$ $B(\lambda) = \left\|\mathbb{E}_{\mu_\lambda}[m] - \bar{\delta m}\right\|_{\mathcal{M}}$ $V(\lambda) = \mathrm{Tr}(\mathrm{Cov}_{\mu_\lambda}(m))$

(2) $\frac{\mathrm{d}}{\mathrm{d}\lambda}(B(\lambda)^2 + V(\lambda))\big|_{\lambda=1}$——

(3) v_c

$$\frac{\mathrm{d}B(\lambda)^2}{\mathrm{d}\lambda}\Big|_{\lambda=1} = -\frac{\mathrm{d}V(\lambda)}{\mathrm{d}\lambda}\Big|_{\lambda=1} \tag{59}$$

"="—— -

7

7.1

SCXSpring SE-1-

(1) ——

(2) -——

(3) ——

(4) $\mathcal{F}_t$——

7.2

——

1: $\lim_{t\to\infty} \mathbb{E}[\|\theta_t - \theta_t^*\|^2] = \infty$ $v_{\min} 10^{-6}$/ 10^9——“” “”

2: $\liminf_{t\to\infty} \mathbb{E}[\|\theta_{t+1}^* - \theta_t^*\|] \geq v_{\min} > 0$ —— 3 1

3: Ornstein-Uhlenbeck$\bar{\theta}_t^*$ $\theta_t^* \theta_t^*$ “”——

4: “”$w(t,s) = 1/t$Spring SE-1 α_t——α_s $\alpha_s \alpha_t \to 0$ “”Spring SE-1——

-

1: **PL** PLv_c PL—— $v\theta_t^*$ PL“”

2: Spring SE-1$\alpha_t \propto 1/t$ “”$\alpha_t\gamma$
“”

3: $b(\lambda)^2 \approx v^2/(\alpha^2\mu^2) \cdot (\lambda/(1-\lambda))^2$ $g_t \approx H(\theta_t - \theta_t^*)$

4: λ^* $\lambda^*(\mu, L, \sigma^2, v)$—— $\lambda^*\hat{v}$ $(v \approx v_c)$——

1: “”least favorable case for detection within the class of distributions with bounded second moment is Gaussian ——$\gamma \cdot \tau = \Omega(1/\delta^2)$

2: δ δ——δ δ δ

3: α
——

$\mathcal{F}_t$

1: **Hilbert** Hilbert TransformerKVLSTM Hilbert

2: A1$\|\mathcal{F}_t(m, \delta m) - \mathcal{F}_t(m', \delta m)\| \leq \rho_t \|m - m'\|$ “” “”

3:
$\frac{\mathrm{d}m}{\mathrm{d}t} = -\gamma(t) \cdot (m - \mathcal{T}(\delta m))$ ——

7.3 Open Problem

Temporal SCX

OP 1 $\rho_t \|\delta m_t\|$“” $\{\mathcal{F}_t\}$

OP 2 “” $I(;\ |\ )$ /—— SCX$I(Y; P \mid S) > 0$

OP 3 ??——
//

OP 4 ——“”—— vs γv “”

OP 5 **-** 2 MLPλMSE MSE$v < v_c v > v_c$

OP 6 A1 —— “”“”“”

7.4 SCX

Temporal SCXSCX

(1) “” “” SCX

(2) **Spring SE-1**Spring SE-1 Spring

(3) **SCX**Spring$\mathcal{M}_t$ λW v——

(4) →- →SCX

7.5

SCX——

1: [] 1–3v SCXθ_t^* $\hat{v}$ θ_s^*—— “”

2: [] Spring$\lambda \approx 0.99$ Yajie$\lambda \approx 0.9$

3: - [] 3$\gamma \cdot \tau$ $\tau\gamma$ $\gamma \leq \gamma_{\max}$ —— “”SPRT

8

—— **SCX**

(1) **1**——

(2) **-2** “”“”

(3) **3** ——

——

——2$\frac{\mathrm{d}\,\mathrm{MSE}}{\mathrm{d}\lambda} = 0$

[] -“-” // ——

SCX2026Spring SE-1

A

??

Proposition A.1. $w(t,s) = 1/W \cdot \mathbf{1}[t - s \leq W]$ *MSE*

$$\mathrm{MSE}(W) = \frac{v^2 W^2}{4\alpha^2\mu^2} + \frac{\alpha\sigma^2}{\mu W} \tag{60}$$

W^*

$$W^* = \left(\frac{2\alpha^3\sigma^2\mu}{v^2}\right)^{1/3} \tag{61}$$

$v \to 0 W^* \to \infty$ $v \to \infty W^* \to 0$ $W^* = 1 v v_c^{FW} = \sqrt{2\alpha^3\sigma^2\mu}$

Proof. **??**W

- $\frac{1}{W}\sum_{k=0}^{W-1}(\theta^*_{t-k}-\theta^*_t)$ $\left\|\frac{1}{W}\sum_{k=0}^{W-1}(\theta^*_{t-k}-\theta^*_t)\right\|^2=\frac{v^2W^2}{4}$
- $W\alpha^2\sigma^2/W$

$\frac{\mathrm{d}\,\mathrm{MSE}(W)}{\mathrm{d}W}=0W^*$ □

$\lambda=1-1/WWN_{\text{eff}}^{\text{EF}}\approx 2W$ $N_{\text{eff}}^{\text{FW}}=W$ “”

B

Proposition B.1. $\mathcal{M}\cong\mathbb{R}^d$ $\mathcal{F}_\lambda(m)=\lambda m+(1-\lambda)\mathcal{T}(\delta m)$

$$\sigma(\mathcal{F}_\lambda)=\{\lambda\}\cup\{\lambda\cdot\sigma_i:\sigma_i\in\sigma(P_{\mathcal{M}}\mathcal{T})\} \tag{62}$$

$P_{\mathcal{M}}\mathcal{M}$ $\rho(\mathcal{F}_\lambda)=\lambda A1$

Proof. $\mathbb{R}^d\mathcal{F}_\lambda$

$$F_\lambda=\lambda I_d+(1-\lambda)T \tag{63}$$

$T\mathcal{T}\mathcal{M}F_\lambda$ $\lambda+(1-\lambda)\tau_i\tau_iT$ $\|T\|\le 1|\tau_i|\le 1|\lambda+(1-\lambda)\tau_i|\le 1$ $\max_i|\lambda+(1-\lambda)\tau_i|=\lambda+(1-\lambda)=1\tau_i=1$ $\lambda|\tau_i|<1$ □

$\tau_i=1\mathcal{T}$ $\mathcal{F}_\lambda 1$—— $\tau_i=0\lambda\lambda$ A1——
[] Transformer “”attention dropout / stochastic depth

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